Principal Component Analysis (PCA)

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- However, the well-known formulation was given by Harold Hotteling in 1933.
- It is also known as *Karhunen-Loève* transformation and it is a non-supervised method.

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- Let's assume that each data point has its specific scale.
- Training a model on such a dataset will definitely be a waste of time.

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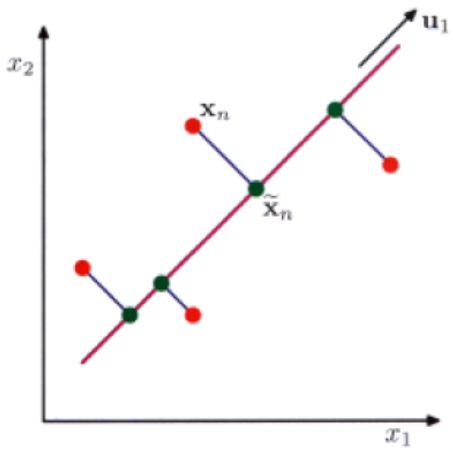

[Bishop2006]

Figure 1: An example of PCA.

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- So, we say that u_1 must be the eigenvectors of C_X, and the variance of the data is given by the eigenvalues.

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$$\frac{1}{N} X^T X u_i = \lambda_i u_i \tag{7}$$

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- Where the $N^{-}1XX^T$ matrix is a square $n \times n$ matrix. Therefore, the cost of calculating the eigenvectors becomes $O(N^3)$ instead of $O(D^3)$.

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- In order to keep only significant information, the $W's$ vectors with the less eigenvalue are rejected and the dimensionality of the original dataset is finally reduced.

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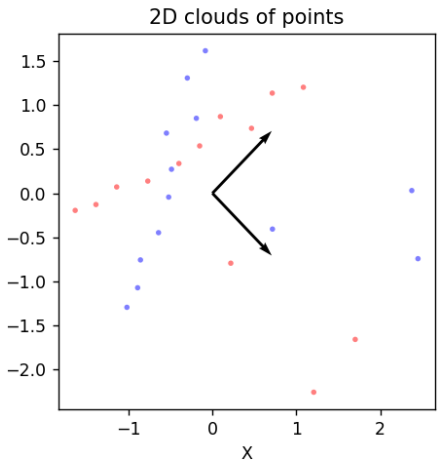

Figure 2: Original and projected points in the same plot.

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Thank you for your attention

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